# Personal Learning Roadmap – Next 90 Days (Post-Internship Development Plan)

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Current Baseline (End of Day 30): Completed the FinDoc-AuditEngine v3.0 working prototype (FastAPI, Pydantic v2, deterministic 3-way reconciliation, regex PII masking, 7-feature synthetic-baseline IsolationForest with perturbation attribution, and PyTest suite).
90-Day Goal: Systematically address the current prototype's limitations (persistence, authentication, real-world ML evaluation/calibration, optical OCR, and asynchronous scaling) while strengthening core Software & AI Engineering fundamentals.

## Month 1 (Days 1–30): Durable Persistence, API Security & Document OCR

| Week | Days | Focus Area | Concrete Learning & Implementation Tasks | Measurable Weekly Target |
| :--- | :--- | :--- | :--- | :--- |
| Week 1 | Days 1–7 | Relational Persistence (PostgreSQL + SQLAlchemy 2.0 + Alembic) | Replace the prototype's in-memory AUDIT_LEDGER Python list with persistent PostgreSQL tables (invoices, purchase_orders, audit_decisions) and schema migrations. | Working async CRUD layer where audit history survives server restarts; verified by integration tests. |
| Week 2 | Days 8–14 | Idempotency, Caching & Concurrency (Redis) | Add Redis-backed idempotency keys (X-Idempotency-Key) and sliding-window velocity counters to detect duplicate or split-invoice submissions. | Benchmark concurrent requests using locust or httpx.AsyncClient with zero duplicate ledger writes. |
| Week 3 | Days 15–21 | API Security (OAuth2 / JWT, RBAC & Rate Limiting) | Implement user authentication and role-based access control (Auditor, Finance_Manager, Director) to secure approval endpoints and ledger exports. | Protected API routes with unit tests verifying 401 Unauthorized and 403 Forbidden role boundaries. |
| Week 4 | Days 22–30 | Optical OCR & Context-Aware PII Redaction (Tesseract / LayoutLM & Presidio) | Upgrade /api/v1/audit/parse-raw from regex-only text parsing to actual PDF/image OCR table extraction, and pair regex PII rules with Named Entity Recognition (NER). | End-to-end sample PDF invoice ingestion test extracting line items and masking names/addresses. |

## Month 2 (Days 31–60): Rigorous ML Evaluation, Calibration, MLOps & Containerization

| Week | Days | Focus Area | Concrete Learning & Implementation Tasks | Measurable Weekly Target |
| :--- | :--- | :--- | :--- | :--- |
| Week 5 | Days 31–37 | Benchmarking on Public Financial Anomaly Datasets & Score Calibration | Evaluate IsolationForest, LocalOutlierFactor, and XGBoost on public labeled fraud/anomaly datasets; study Platt scaling and isotonic regression for probability calibration. | Reproducible evaluation notebook reporting ROC-AUC, PR-AUC, and Brier score calibration curves. |
| Week 6 | Days 38–44 | Experiment Tracking & Model Registry (MLflow) | Log model hyperparameters, StandardScaler artifacts, feature medians, and evaluation metrics in MLflow instead of fitting on synthetic data at app startup. | Versioned model artifact loaded deterministically from an MLflow registry path at startup. |
| Week 7 | Days 45–52 | Data Drift Monitoring (EvidentlyAI & Population Stability Index) | Build a scheduled drift evaluation script that computes PSI and Kolmogorov-Smirnov statistics across the 7 input features over time. | Automated HTML/JSON drift report flagging simulated distribution shifts ($\text{PSI} > 0.20$). |
| Week 8 | Days 53–60 | Code Coverage (pytest-cov), Docker Multi-Stage Builds & CI Gates | Measure actual line and branch coverage using pytest-cov, containerize the FastAPI + PostgreSQL stack with docker-compose, and enforce coverage checks in GitHub Actions. | Verified pytest --cov=app report (>= 85% measured line coverage) running automatically on every PR. |

## Month 3 (Days 61–90): Hybrid RAG, Asynchronous Workflows & Interview Readiness

| Week | Days | Focus Area | Concrete Learning & Implementation Tasks | Measurable Weekly Target |
| :--- | :--- | :--- | :--- | :--- |
| Week 9 | Days 61–67 | Hybrid Retrieval (BM25 + Dense Vector Search) & Evaluation (RAGAS) | Combine keyword search (BM25) with vector embeddings (Qdrant / pgvector) to retrieve procurement policy clauses for flagged invoices; measure faithfulness. | Policy-grounded explanation module evaluated on a 30-question test set. |
| Week 10 | Days 68–75 | Asynchronous Task Queues (Celery / RabbitMQ) | Offload batch invoice processing and report generation to background worker queues with retry and dead-letter handling. | Async batch audit endpoint with job-status polling (PENDING -> COMPLETED). |
| Week 11 | Days 76–82 | Data Structures, Algorithms & Backend System Design | Practice 45+ medium/hard DSA problems (graphs, heaps, dynamic programming) and review 4 core system design architectures. | Documented system design notes and timed coding problem solutions. |
| Week 12 | Days 83–90 | Portfolio Consolidation & Mock Technical Interviews | Conduct full mock technical interviews covering all projects (FinDoc-AuditEngine, LEGITIFY, CampusLLM, FinTrack, Solar Forecasting), refine documentation, and finalize applications. | Clean, reproducible GitHub portfolio with verified READMEs and architecture guides. |